1 · BUSINESS PROBLEM

SmartCart Customer Segmentation

From one-size-fits-all marketing to data-driven personas

Business problem

- Wasted marketing budget
- VIP churn
- Low conversion

ML translation

- Unsupervised clustering
- 2,240 customers, 22 attributes

Outcome

- 4 customer personas
- Strategy, KPIs, A/B plan

2 · DATA UNDERSTANDING

Understanding the data with Python statistics

$666,666
max income
(median $51,382)

24
missing incomes (1.07%)

5.79 | 4.08 | 2.66
store | web | catalog purchases

14.9%
campaign response rate

Skewed income	→	Median imputation, outlier flag
Six spend columns	→	One Total_spending feature
Mixed scales	→	StandardScaler

Python · pandas · scikit-learn

3 · CLEANING & FEATURE ENGINEERING

Feature engineering in Python

Age

2026 − Year_Birth

Tenure days

Latest Dt_Customer − own Dt_Customer

Total_spending

Sum of 6 product-spend columns

Total_childrens

Kidhome + Teenhome

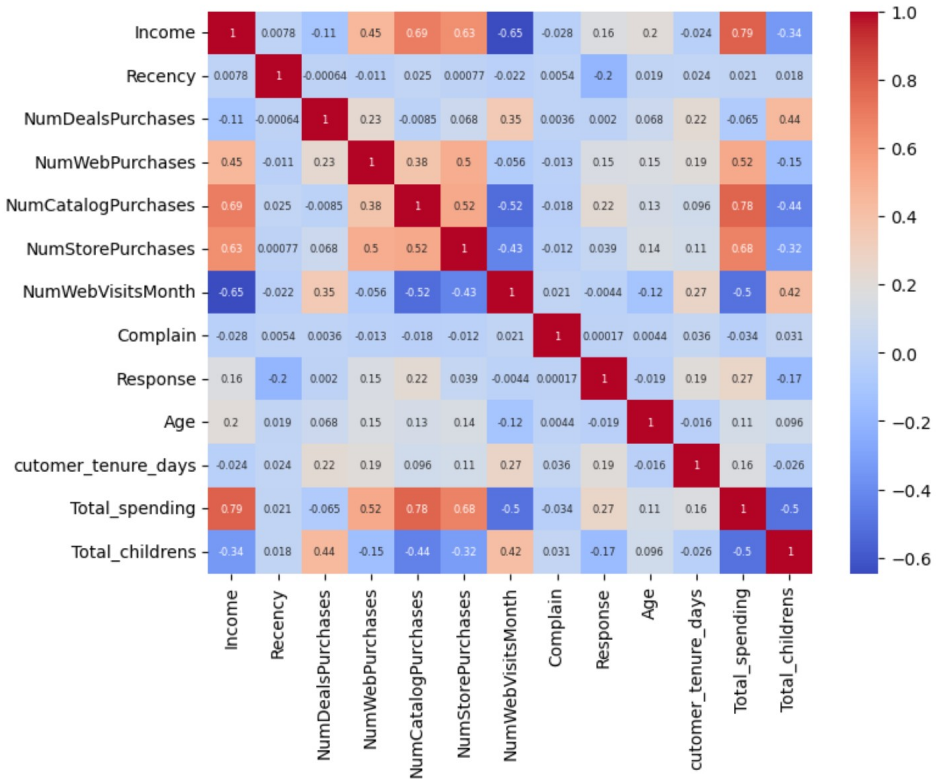
22 → 15

features

- 24 nulls: median imputation
- Education 5 → 3, marital 8 → 2
- 12 columns dropped

4 · EDA, ENCODING & PCA

EDA finds the features that matter



Outliers

Removed income > $600k and age > 90

Heatmap

Income ↔ spend r ≈ 0.79; children ↔ spend r ≈ −0.50

Encode, scale, PCA

One-hot + StandardScaler; PCA 18D → 2D / 3D

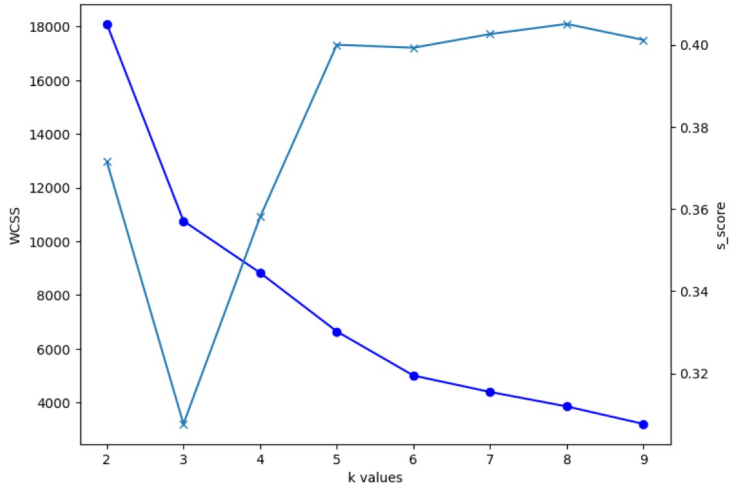
Why it matters

Stable model inputs, drivers marketing can explain

5 · K-VALUE & ALGORITHM CHOICE

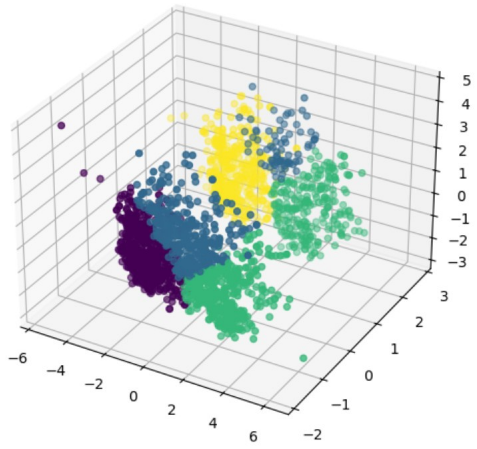
Model Selection: Unsupervised ML Algorithms

Elbow (WCSS) + Silhouette



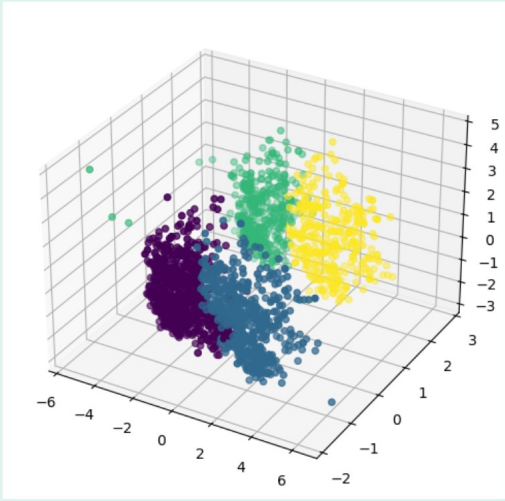
Elbow bends at K = 4; silhouette ≈ 0.36 at K = 4

K-Means (K=4)



Fast centroid baseline; assumes spherical clusters

Agglomerative (final)

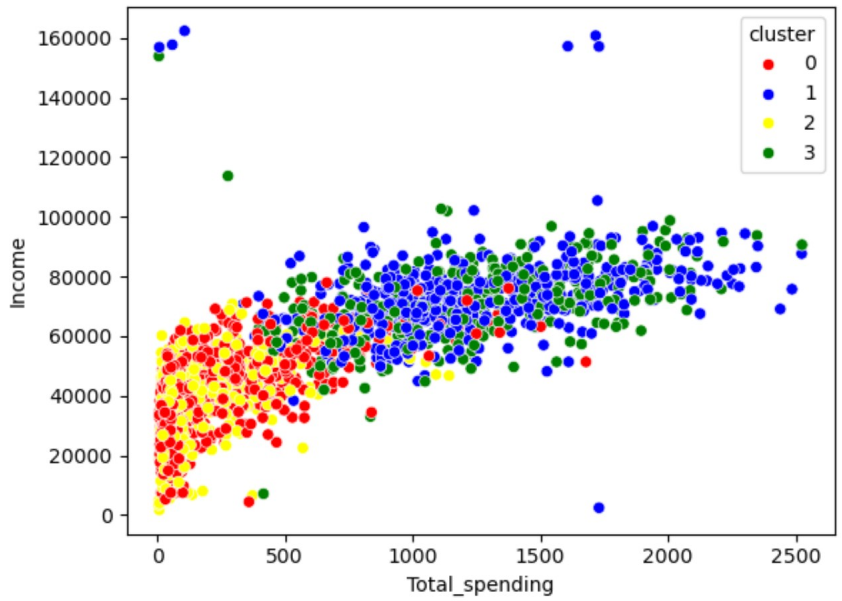


Ward hierarchy: tighter, better-separated clusters

6 · CLUSTER SUMMARY

Four personas, four strategies

Income vs total spending, by cluster



Cluster 0

Family Shoppers

Browse a lot, buy little

Convert with bundles

Cluster 1

Premium Couples

High spend, all channels

Protect with loyalty perks

Cluster 2

Digital Browsers

Low income, many visits

Convert with flash sales

Cluster 3

Premium Singles

Top spend, best response

High-touch, high-ROI offers

7 · RECOMMENDATIONS & DEPLOYMENT

Targeted actions, then rollout

Family Shoppers	Coupons and family bundles	Conversion, AOV
Premium Couples	VIP loyalty and early access	Retention, LTV
Digital Browsers	Flash sales, cart-abandonment offers	Visits to purchases
Premium Singles	1-on-1 offers, concierge	Campaign ROI

Step 1
Pipeline
Assign new customers to clusters in the warehouse

Step 2
CRM
Push labels to Klaviyo or Salesforce

Step 3
A/B test
Targeted vs generic; track lift, LTV, CAC

Step 4
Monitor
Re-cluster quarterly